

2.7 Zen3: The AMD 2nd-Generation 7nm x86-64 Microprocessor Core

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“Zen 3” is the first major microarchitectural redesign in the AMD Zen family of microprocessors. Given the same 7nm process technology as the prior-generation “Zen 2” core [1], as well as the same platform infrastructure, the primary “Zen 3” design goals were to provide: 1) a significant instruction-per-cycle (IPC) uplift, 2) a substantial frequency uplift, and 3) continued improvement in power efficiency. The core complex unit (CCX) consists of 8 “Zen 3” cores, each with a 0.5MB private L2 cache, and a 32MB shared L3 cache. Increasing this from 4 cores and 16MB L3 in the prior generation provides additional performance uplift, in addition to the IPC and frequency improvements. The “Zen 3” CCX shown in Fig. 2.7.1 contains 4.08B transistors in 68mm², and is used across a broad array of client, server, and embedded market segments.

The “Zen 3” design has several significant microarchitectural improvements. A high-level block diagram is shown in Fig. 2.7.2. The front-end had the largest number of changes, including a 2× larger L1 Branch-Target-Buffer (BTB) at 1024 entries, improved branch predictor bandwidth by removing the pipeline bubble on predicted branches and faster recovery from mispredicted branches, and faster sequencing of op-cache fetches. In the execution core, the integer unit issue width was increased from 7 to 10, including dedicated branch and store pipes, the reorder buffer was increased by 32 entries to 256, while in the floating-point unit, the issue width was increased from 4 to 6 and the FMA latency was reduced from 5 to 4 cycles. In the load-store unit, both the maximum load and store bandwidth was increased by one to 3 and 2, respectively, and the translation look-aside buffer (TLB) was enhanced with 4 additional table walkers for a total of 6. Overall, the “Zen 3” core delivers +19% average IPC uplift over “Zen 2” across a range of 25 single-threaded industry benchmarks and gaming applications, with some games showing greater than +30% [2].

To support the change to an 8-core CCX for “Zen 3,” and the corresponding increase in L3 cache slices from 4 to 8, the cache communication network was converted to a new bi-directional ring bus with 32B in each direction, providing both high bandwidth and low latency. The 8-core CCX with 32MB L3 provides 2× the available local L3 cache as “Zen 2” per core, providing significant uplift on many lightly-threaded applications. The L3 cache also contains the through-silicon vias (TSVs) to support AMD V-Cache, allowing an additional 64MB AMD 3D V-Cache to attach to the base die via direct copper-to-copper bond, which can triple the L3 capacity per CCX to 96MB. Another key change from “Zen 2” was converting the L3 cache from the high-current (HC) bitcell to the high-density (HD) bitcell, yielding an area improvement of 14%, and a leakage reduction of 24% of the cache array, all while matching the higher core clock frequencies.

The critical physical design challenge of the new “Zen 3” cache was the implementation of the ring bus and the interface to the AMD V-Cache. As can be seen in Fig. 2.7.3, two columns of TSVs run down the left and right halves of the cache, providing connectivity to the stacked AMD V-Cache that effectively triples the memory capacity of each slice. The TSV interface supports >2Tb/s per slice, for an aggregate bandwidth between the two dies of >2TB/s. The ring bus has a cross-sectional bandwidth of >2Tb/s to match the peak core-L3 bandwidth. Architecturally, the L3 cache supports up to 64 outstanding misses from each core to the L3, and 192 outstanding misses from L3 to external memory.

“Zen 3” uses the same 13-metal telescoping stack as “Zen 2,” optimized for density at the lower layers and for speed on the upper layers [1]. A key goal of “Zen 3” was to increase frequency at fixed voltage by 4% across the full range of voltage, with 6% increase at high-voltage to drive a corresponding improvement in single-thread performance. Median silicon measurements of frequency vs. voltage are shown in Fig. 2.7.4, demonstrating success in meeting the goal, and more importantly, the ability to achieve superior single-thread operation at 4.9GHz. Structured logic placement, judicious cell selection, targeted use of low V_Ts, and wire engineering were used to drive the large frequency increase at high voltage, and the tiles of the core were kept small to enable multiple design iterations per week. To deliver the average IPC uplift of 19% and frequency increase of 6%, the CCX effective switched capacitance (Cac) increased by 15%. Maintaining a ratio of $\Delta\text{Cac}/\Delta\text{IPC}$ at less than one produces a more power efficient core. A breakdown of the “Zen 3” Cac is shown in Fig. 2.7.5. The fraction of Cac consumed by clock gates has increased slightly over the previous generation and the fraction consumed by flops and combinational logic has decreased slightly due to the additional effort to improve the clock gating efficiency of “Zen 3.” A modest increase in leakage power was outweighed by the frequency increase to further improve power efficiency. While the Cac and leakage increase leads to higher power at fixed voltage, at ISO performance the “Zen 3” core delivers up to 20% higher performance/W, as shown in Fig. 2.7.6.

The “Zen 3” core IP was simultaneously used in two distinct tape-outs. The first was the “Zen 3” core complex die (CCD) chiplet, comprising the CCX, a system management unit (SMU), and an Infinity Fabric On-Package (IFOP) SerDes link to connect to a separate IO die (IOD). The primary upgrade from the prior “Zen 2” CCD [3] was replacing the two 4-core “Zen 2” CCXs with a single 8-core “Zen 3” CCX, which allowed for a more streamlined on-die Infinity Fabric to connect only one CCX and leveraged the remaining IP from the prior-generation CCD. The size of this CCD is 81mm², containing 4.15B transistors. Similar to the prior generation, the CCD was combined with a low-cost 12nm IOD to provide very cost-effective performance, and the modularity of chiplets enabled product configurations to service the entire breadth of the server and desktop PC markets. AMD “Milan” server products combine the server IOD with 2-to-8 “Zen 3” chiplets to deliver a cost-optimized product stack from 16-to-64 cores. AMD “Vermeer” client products combine the much smaller client IOD with 1-to-2 “Zen 3” chiplets to cover the spectrum from top-end 16-core performance desktop products to mainstream 8-core value products.

The second tape-out was AMD “Cezanne”, a monolithic APU, which upgraded the prior generation APU [4] from two “Zen 2” CCXs to a single “Zen 3” CCX, having the same core as in the CCD and a cut-down 16MB L3 Cache. This die, measuring 180mm² with 10.7B transistors, contains an 8-compute-unit Vega graphics engine, a multimedia engine, a display engine, an audio engine, two DDR4/LPDDR4 memory channels, as well as PCIe, USB and SATA ports.

Combining both the IPC and frequency gains of “Zen 3” over “Zen 2” yields significant performance gains in both single- and multi-thread performance across the entire spectrum of product segments. Figure 2.7.7 shows the Cinebench R20 benchmark score increase of single-thread (1T) performance of 17% on the “Vermeer” desktop part: 13% from IPC and 4% from increased frequency. The multi-threaded Cinebench R20 improvement on the “Cezanne” mobile APU shows 14% uplift. Despite being constrained to the same process technology as the prior generation, the “Zen 3” family of products deliver a substantial boost in performance over the prior generation through the combination of a ground-up redesign of the underlying Zen microarchitecture and the innovative work of the physical design team to achieve up to 6% higher frequency at the same voltage.

References:

- [1] T. Singh et al., “Zen 2: The AMD 7nm Energy-Efficient High-Performance x86-64 Microprocessor Core,” *ISSCC*, pp. 42-43, 2020.
- [2] M. Evers et al., “Next Generation ‘Zen 3’ Core,” *IEEE Hot Chips Symp.*, 2021.
- [3] S. Naffziger et al., “AMD Chiplet Architecture for High-Performance Server and Desktop Products,” *ISSCC*, pp. 44-45, 2020.
- [4] S. Arora et al., “AMD Next Generation 7nm Ryzen 4000 APU ‘Renoir’,” *IEEE Hot Chips Symp.*, 2020.

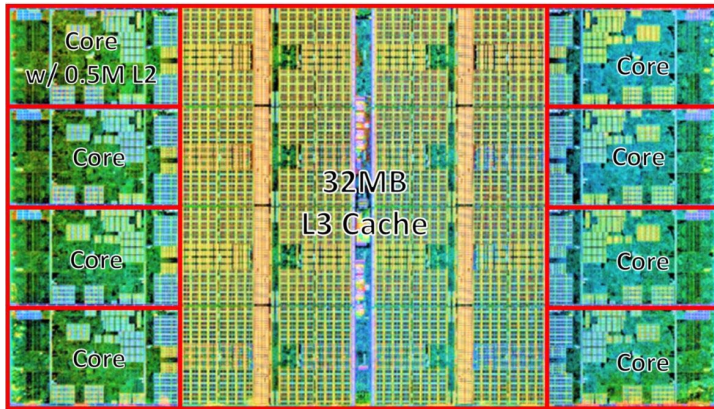


Figure 2.7.1: Die photo of "Zen 3" core complex unit (CCX).

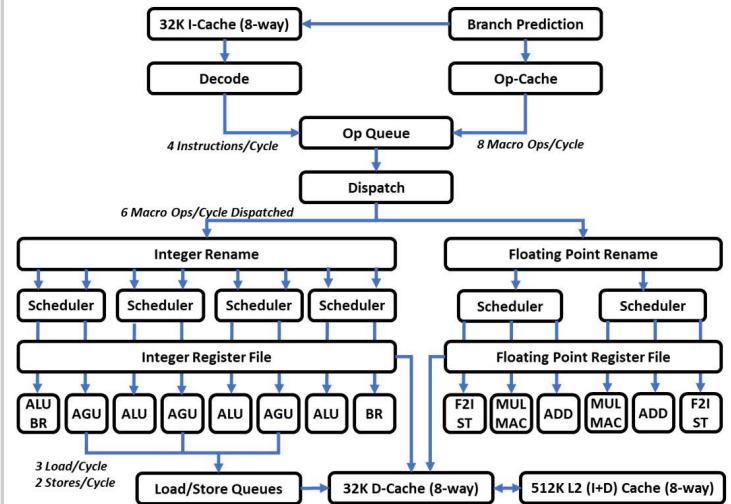


Figure 2.7.2: "Zen 3" architecture.

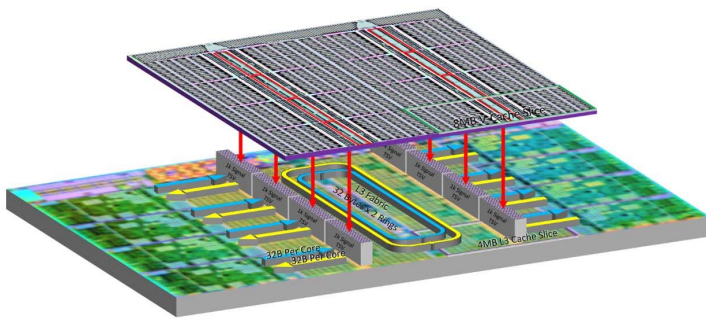


Figure 2.7.3: L3 ring bus design.

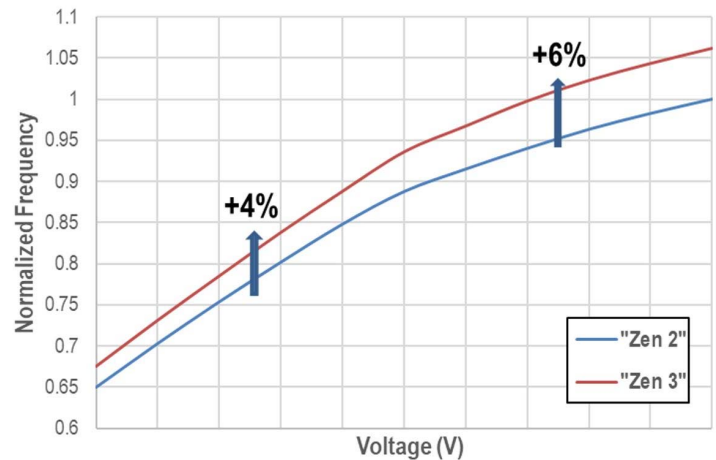


Figure 2.7.4: Frequency improvement.

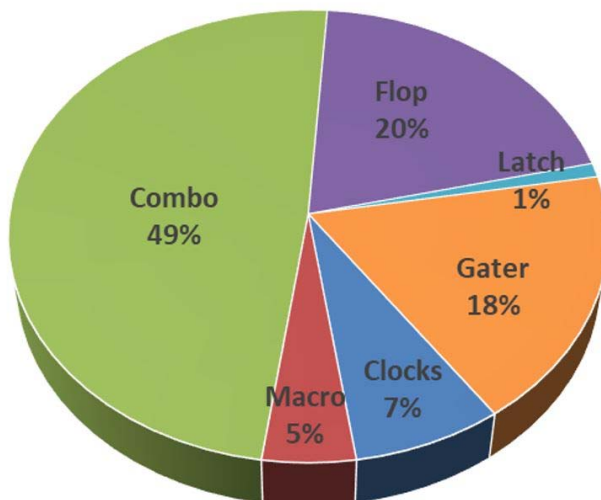


Figure 2.7.5: Cac breakdown.

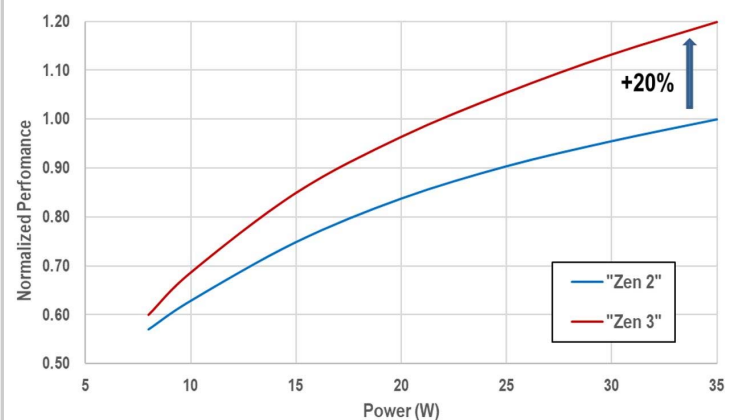
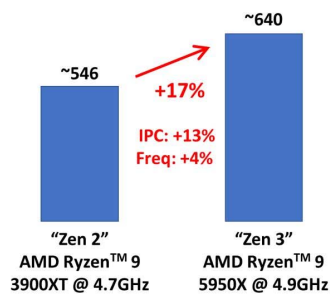


Figure 2.7.6: Performance (average) versus power for one 8C32M "Zen 3" CCX versus two 4C16M "Zen 2" CCXs.

Performance Desktop
Cinebench R20 1T Performance



Premium 15W 8-Core Mobile
Cinebench R20 nT Performance

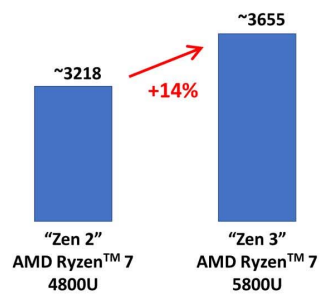


Figure 2.7.7: Cinebench R20 performance improvement.